ESPRIT · 2026

FINAL YEAR PROJECT

PEAXIS

A Modular AI-Powered Hiring Operating System

BIBANI Mohamed Habib Allah

École Supérieure Privée d'Ingénierie et de Technologie — ESPRIT

ACADEMIC SUPERVISOR
Mme Olfa Mannai

COMPANY SUPERVISOR
Mr. Fedi Naimi

1 INTRODUCTION & CONTEXT

Presentation Overview

2 INDUSTRY & PROBLEM

Recruitment in the Digital Era

- Application volume
- Recruiter workload
- Candidate expectations
- AI adoption in HR

2 COMPANY PRESENTATION

Prospecter — Internship Host Company

AI-powered B2B platform for outbound prospecting.

MISSION

Autonomous AI SDRs help B2B teams scale outbound prospecting.

AI / LLM · B2B SAAS · SALES AUTOMATION · OUTBOUND GTM

AI SDR Engine

Lead Intelligence

Outbound Automation

Pipeline Analytics

Strategic context — LLM workflows · Multi-tenancy · Background workers

3 INTERNSHIP

Internship Context

Software Engineering internship at **Prospecter**, an AI-powered B2B platform.

Full-Stack Engineering

AI Integration

Platform Engineering

Scope: Independent product · Production-inspired engineering

SOFTWARE ENGINEER

AI INTEGRATION

SOLO CONTRIBUTOR

2 INDUSTRY & PROBLEM

Current recruitment challenges

42 days
Average time-to-hire
SHRM 2024

75%
Recruiters overloaded
LinkedIn Talent Report

60%
Candidates report no feedback
Indeed Survey 2023

Manual screening

Keyword-based ATS

Inconsistent evaluation

Candidate experience

Cost of a bad hire: averages **$14.9K**; strong candidates can accept competing offers within days.

Engineering response: keep screening responsive, make every score reviewable, and keep candidates informed.

4 EXISTING SOLUTIONS & GAP

Competitive **Analysis**

How existing recruitment platforms cover core ATS, candidate experience, and explainability.

CAPABILITY	GREENHOUSE	LEVER	WORKABLE	ASHBY	HIREVUE	EIGHTFOLD AI	PEAXIS
Core ATS	✓	✓	✓	✓	✗	partial	✓
Candidate Portal	basic	basic	basic	✗	✓	✗	✓
Explainable AI	✗	✗	partial	✗	partial	✓	✓
Integrated Ecosystem	✗	✗	partial	partial	✗	✗	✓
Tenant Isolation	✓	✓	✓	✓	✓	✓	✓

Observation: few platforms combine core ATS workflows, a candidate-facing portal, and explainable AI scoring in a single, SME-accessible product.

4 EXISTING SOLUTIONS & GAP

The gap: integrated, explainable AI

AI black boxes

Fragmented tooling

No candidate intelligence

Evidence first

Human authority

Durable operations

This gap defines the project's objective
One auditable recruitment platform.

5 METHODOLOGY & REQUIREMENTS

Engineering Methodology

SPRINT CYCLE

- 1 **Backlog**
- 2 **Sprint Planning**
- 3 **Build & Test**
- 4 **Review & Improve**

HOW WORK WAS MANAGED

- **Scrum Sprints**
- **Kanban Board**
- **Incremental Delivery**
- **Sprint Review**

5 FUNCTIONAL REQUIREMENTS

Core platform requirements

Core platform capabilities.

FR-01
Authentication & Security
- Argon2 password protection
- JWT + OAuth sign-in

FR-02
Multi-Tenant Architecture
- Business-scoped records
- Tenant context enforced

FR-03
Roles & Permissions
- Role and plan guards
- Feature access by subscription

FR-04
Job Management
- Create and filter jobs
- Hybrid text + vector search

FR-05
Application Pipeline
- Tracked application stages
- Async assessment updates

5 FUNCTIONAL REQUIREMENTS

AI requirements

Implemented intelligence capabilities.

FR-06
Evidence Matching
- Requirement-level assessment
- 0–100 alignment score

FR-07
CV Parsing
- PDF / DOCX / TXT upload
- Async structured extraction

FR-08
Candidate Onboarding
- Apply with a CV
- Confirm parsed profile

FR-09
Explainable Review
- Cited strengths and gaps
- Verification state per requirement

FR-10
Reliable AI Processing
- Queued AI work items
- Retry and failure status

5 NON-FUNCTIONAL REQUIREMENTS

Quality **attributes**

Reliability, security, and maintainability.

Container-ready

NFR-01 Deployment Readiness

Web, API, worker, AI runtime, and data services are separated.

Implemented controls

NFR-02 Security

Auth, file gates, rate limits, tenant scoping, and service secrets.

No AI in request path

NFR-03 Responsive Requests

Parsing and assessment run outside the HTTP request path.

Explicit work states

NFR-04 Failure Handling

Retries and durable work states expose processing failures.

Cited assessment trail

NFR-05 Explainability

Scores retain evidence, gaps, and verification state.

Modular platform

NFR-06 Modularity

Separate web apps, API, worker, and inference boundary.

6 PROPOSED SOLUTION

PEAXIS — Modular AI Hiring Operating System

Business foundation
PEAXIS Core

Recruiter workspace
PEAXIS Hire

Candidate portal
PEAXIS Jobs

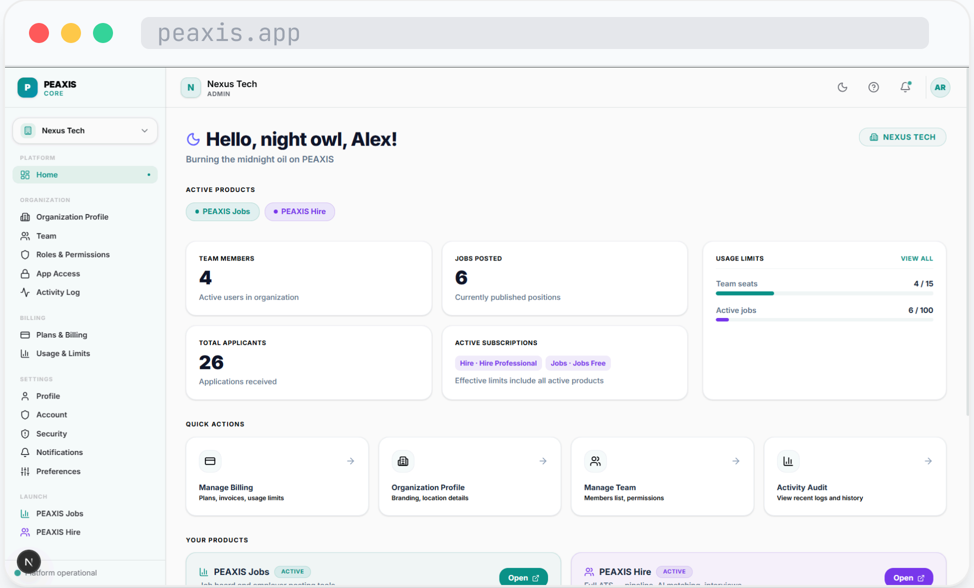
BUSINESS ADMIN
PEAXIS Core

RECRUITER / HR TEAM
PEAXIS Hire

CANDIDATE / TALENT
PEAXIS Jobs

6 PROPOSED SOLUTION

PEAXIS Core



PEAXIS Core — access, business context, and plan controls

Identity & Access

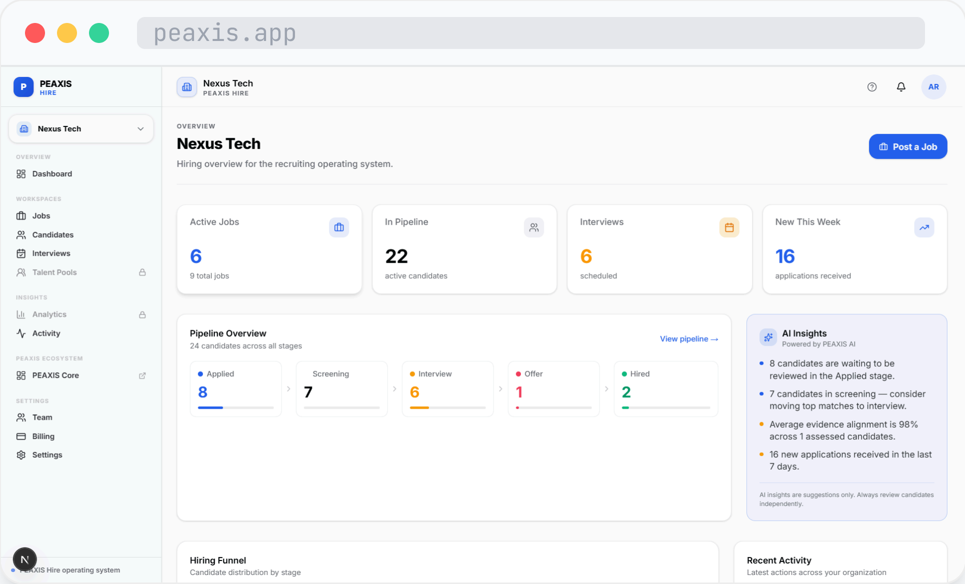
Plans & Entitlements

Business Context

One control centre for every module

6 PROPOSED SOLUTION

PEAXIS Hire



PEAXIS Hire — recruiter pipeline and assessment workspace

Pipeline workspace

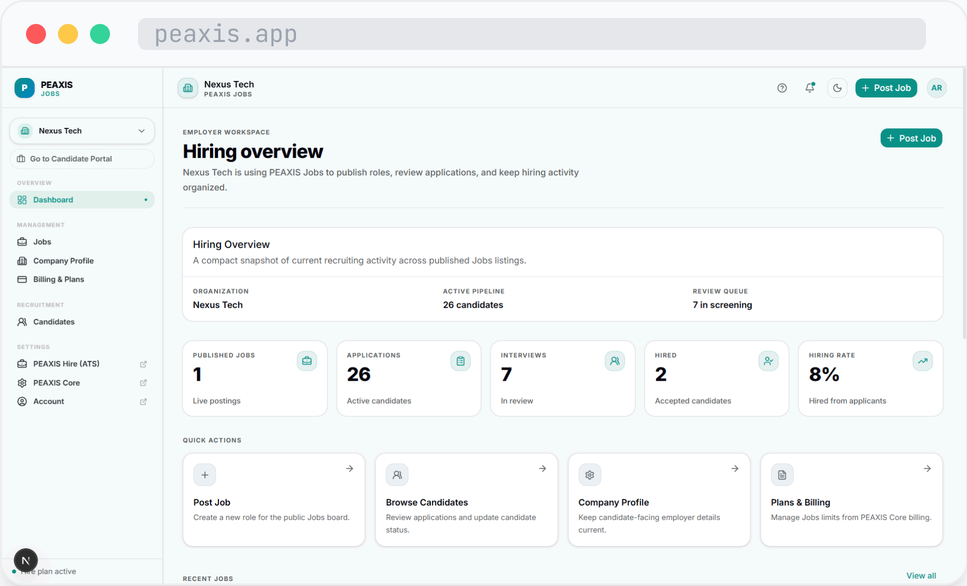
Evidence assessment

Async processing

Pipeline → evidence → recruiter decision

6 PROPOSED SOLUTION

PEAXIS Jobs



PEAXIS Jobs — job discovery and candidate application portal

Job Discovery

Application UX

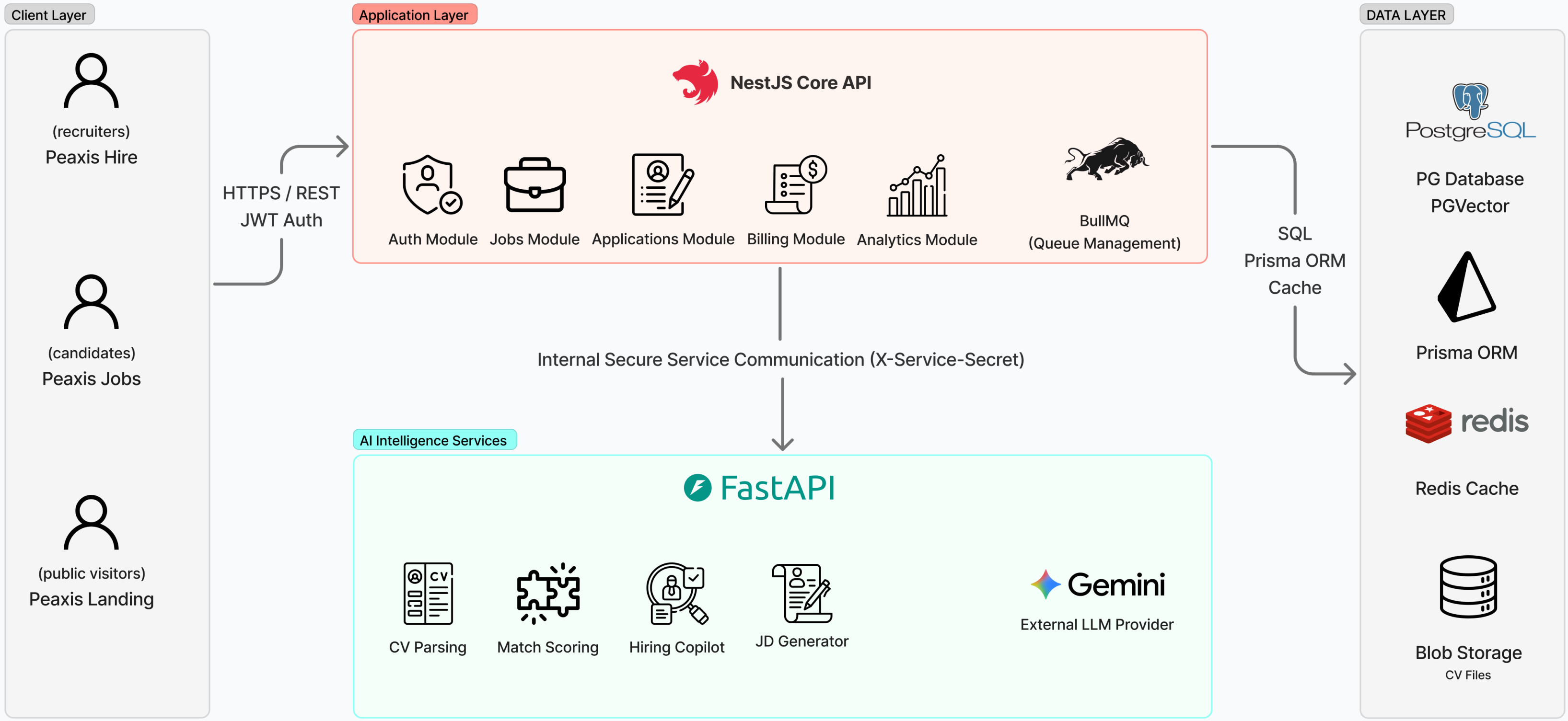
Application Progress

Discover → apply → follow progress

7 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

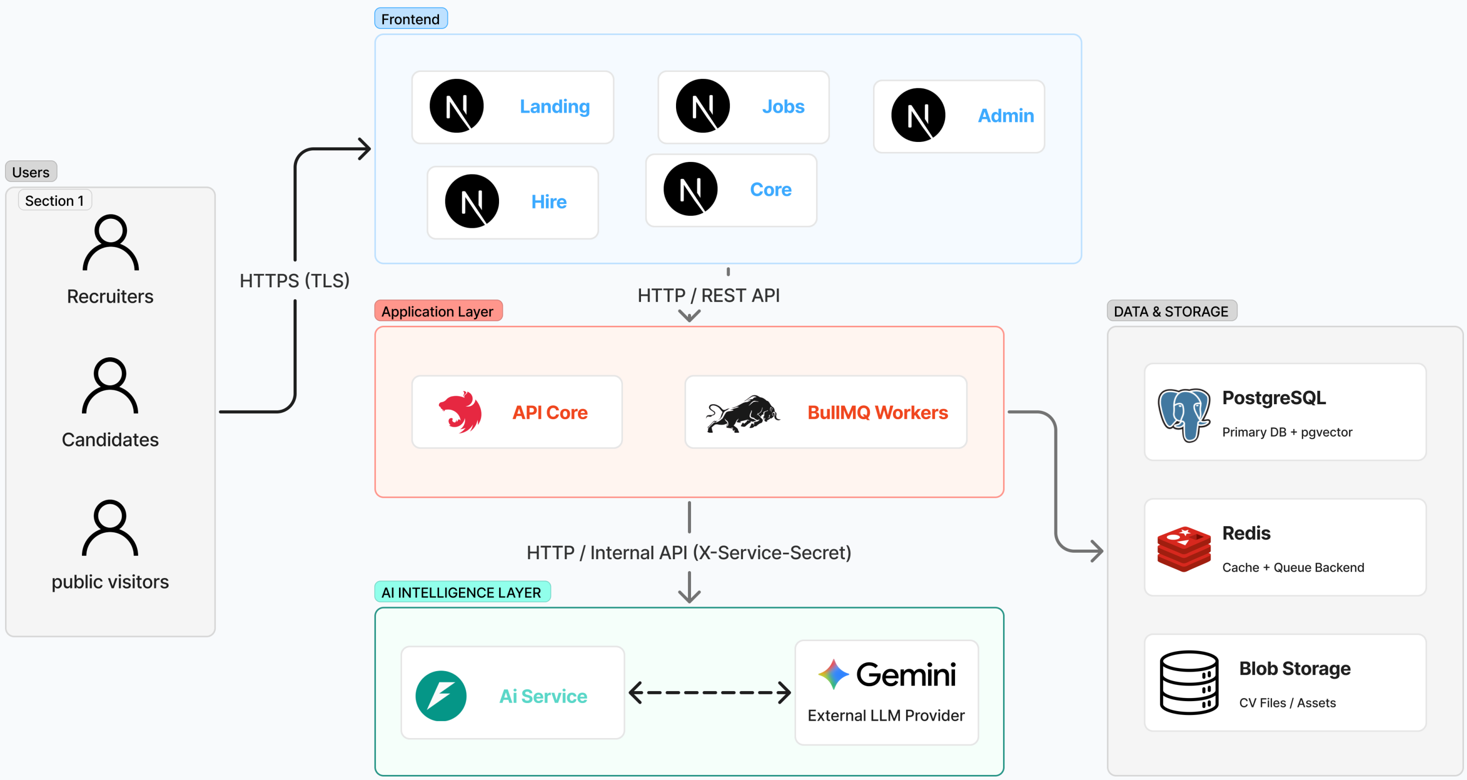
Client, platform, AI, and data boundaries. Gemini or Azure OpenAI is configured at startup.



7 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

Deployment topology for web apps, platform services, AI inference, and storage. Gemini or Azure OpenAI is configured at startup.



8 ENGINEERING DEEP DIVE

AI Runtime Architecture

NestJS orchestrates, FastAPI infers, and PostgreSQL is the source of truth.

Next.js	→	NestJS API	→	AiWorkItem	→	BullMQ	→	Worker	→	FastAPI	→	Provider
user action		policy + write		durable state		ai-tasks queue		execution		inference only		Gemini or Azure

Synchronous request path

Validate, authorize, persist business state, and return a processing response. Short generation or extraction helpers can call FastAPI directly.

Asynchronous worker path

Call FastAPI, persist parse/embedding/assessment outputs, and update the durable work and application state.

Return path: the worker writes results to PostgreSQL; Redis is used for queues, locks, heartbeats, and caches—not authoritative AI state.

8 ENGINEERING DEEP DIVE

CV Parsing Pipeline

A worker turns a validated CV into grounded, structured evidence — without blocking the candidate.

1. INTAKE AND DURABLE PARSING

Candidate upload — public multipart request → **File gate** — type, size, signature, checksum → **ResumeParse** — pending + PARSE_CV work → **Worker + FastAPI** — extract and parse → **Parse result** — READY or NEEDS_REVIEW

Extract text
file layer

PDF, DOCX, and TXT use native extraction. Low-quality PDFs can use Google Vision OCR when enabled.

Structured AI
inference

FastAPI calls the active Gemini or Azure provider at temperature 0 to return strict JSON: skills, experience, education, and citations.

Ground + persist
control

Validation rejects unsupported facts. ResumeParse stores status, warnings, model metadata, and the grounded result; checksum/cache avoid duplicate work.

2. CANDIDATE CONFIRMATION AND RETRIEVAL PREPARATION

Candidate confirms — one-time claim → **candidate.resumeParsed** — structured profile → **Evidence claims** — source-cited facts → **Evidence chunks** — retrieval units → **Embeddings** — vector persistence

8 ENGINEERING DEEP DIVE

Evidence-Based Matching Engine

Exact rules first; pgvector retrieval only when needed; the backend always owns the final decision.

Requirements
recruiter-confirmed definitions

→

Candidate claims
structured résumé facts

→

Evidence assessment
fingerprinted durable work

→

Requirement evaluation
satisfaction + verification

→

Alignment score
ranking uses current score

1. Exact evidence

rules first

NestJS normalizes requirement text, matches candidate claims, and calculates relevant experience months from CV dates.

2. Semantic fallback

retrieval only

If direct evidence is absent, FastAPI creates a 768-dimension embedding; pgvector retrieves up to 8 candidate chunks for classification.

3. Deterministic decision

human authority

NestJS sets satisfaction, verification, requirement weights, and the 0-100 score. The recruiter reviews the result and decides.

The current alignment and ranking score are the same persisted calculation—not a separate opaque candidate similarity model.

8 ENGINEERING DEEP DIVE

How Matching Is Computed

A deterministic evidence algorithm; vectors and LLMs only retrieve or classify citations.

1. Find candidate evidence

Canonical token and simple stem matching compare each requirement with source-cited claims. The fallback searches displayable résumé summary, headline, experience, and project text. Experience requirements merge overlapping date intervals into relevant months.

2. Use vectors only when needed

For unresolved COMPETENCY, RESPONSIBILITY, or TOOL_SYSTEM requirements, FastAPI embeds the requirement. pgvector retrieves only this candidate's chunks at distance ≤ 0.55 (maximum 8). The classifier sees at most 12 claims plus retrieved chunks and may cite only one supplied source.

3. Apply deterministic requirement states

No evidence, mandatory
REQUIRES_VERIFICATION

No evidence, preferred
UNKNOWN

Evidence but years unmet
PARTIALLY_SATISFIED

Evidence + policy satisfied
SATISFIED

4. Persist explainable results

An **EvidenceAssessment** stores input snapshots and the score. Each **RequirementEvaluation** retains satisfaction, constraints, and positive cited evidence.

CURRENT GENERAL:V1 SCORE

score = round(Σ(weight × state value × evidence credit) / Σ(weight))

required 0.55 · experience 0.25 · preferred 0.10 · TRANSFERABLE × 0.60

State value: SATISFIED = 100, PARTIALLY_SATISFIED = 60, all other states = 0. The current ranking score equals this alignment score; no separate opaque cosine ranking is used.

8 ENGINEERING DEEP DIVE

AI Models by Use Case

Gemini extracts or classifies evidence; deterministic backend rules calculate the final score.

Use case	Technology	Role
CV parsing	Gemini 3.5 Flash	Strict structured résumé JSON
Evidence classification	Gemini 3.5 Flash	Classify supplied citations only
Embeddings	gemini-embedding-001	768D semantic retrieval with pgvector
OCR fallback	Google Vision OCR	Recover text from poor PDFs
Final score	NestJS + PostgreSQL	Deterministic 0–100 calculation

Gemini-only model path

FastAPI isolates inference

LLMs do not rank candidates

8 ENGINEERING DEEP DIVE

Performance Optimizations

Bounded retrieval, cache reuse, and durable work reduce synchronous AI latency.

Redis cache

Bounded parse, embedding, and selected-generation reuse; never the source of truth.

Durable worker

Slow inference happens outside HTTP requests with task-specific retry policies.

Hybrid job search

PostgreSQL full-text search merges with pgvector semantic results.

Chunk retrieval

Evidence retrieval is constrained to the candidate and returns only a small cited set.

Provider retries

The inference client retries transient provider/network failures.

Graceful search

If query embedding fails, public job search returns keyword results with degraded status.

Known limit: pgvector is enabled, but the baseline migration does not create an ANN vector index.

8 ENGINEERING DEEP DIVE

Technical Challenges & Solutions

Safeguards and limitations remain explicit for technical review.

Challenge	Solution	Outcome
AI latency	Durable work + worker retries	Responsive request path
CV reliability	File gates + quality gate + optional OCR	READY or NEEDS_REVIEW state
Grounding	Citation-bound evidence classification	No free-form evidence invention
Explainability	Snapshots, evaluations, and citations	Auditable assessment trail
Provider outage	Retries and explicit failure states	Recoverable, observable processing
Known access gap	Candidate recommendation visibility needs remediation	Documented before production

9 ROADMAP

Roadmap & Future Work

Next: harden the delivered system before widening AI capability.

NEAR-TERM

Access control

Enforce active membership for all AI-generated content and correct recommendation visibility.

Atomic delivery

Commit application and assessment work atomically with a transactional outbox pattern.

Reviewer policy

Make overrides affect an explicit versioned assessment policy and explainability.

MID-TERM

FinOps coverage

Route every chargeable AI call through operations and credit ledger reconciliation.

Prompt controls

Centralize prompt construction and test every user-controlled input surface.

Vector lifecycle

Add stale-chunk cleanup and measure before introducing an ANN vector index.

LONG-TERM VISION

Governed expansion

Expose only supported FastAPI helpers through tenant-authorized NestJS contracts.

Data governance

Define provider retention, PII classification, consent, and operational redaction.

10 RESULTS & CONCLUSION

Results & Conclusion

Engineering Achievements

- Full-stack recruitment platform

Architectural Achievements

- Controlled AI architecture

Lessons Learned

- Human authority stays central

Evidence, not opaque ranking.

Thank you

Questions?

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